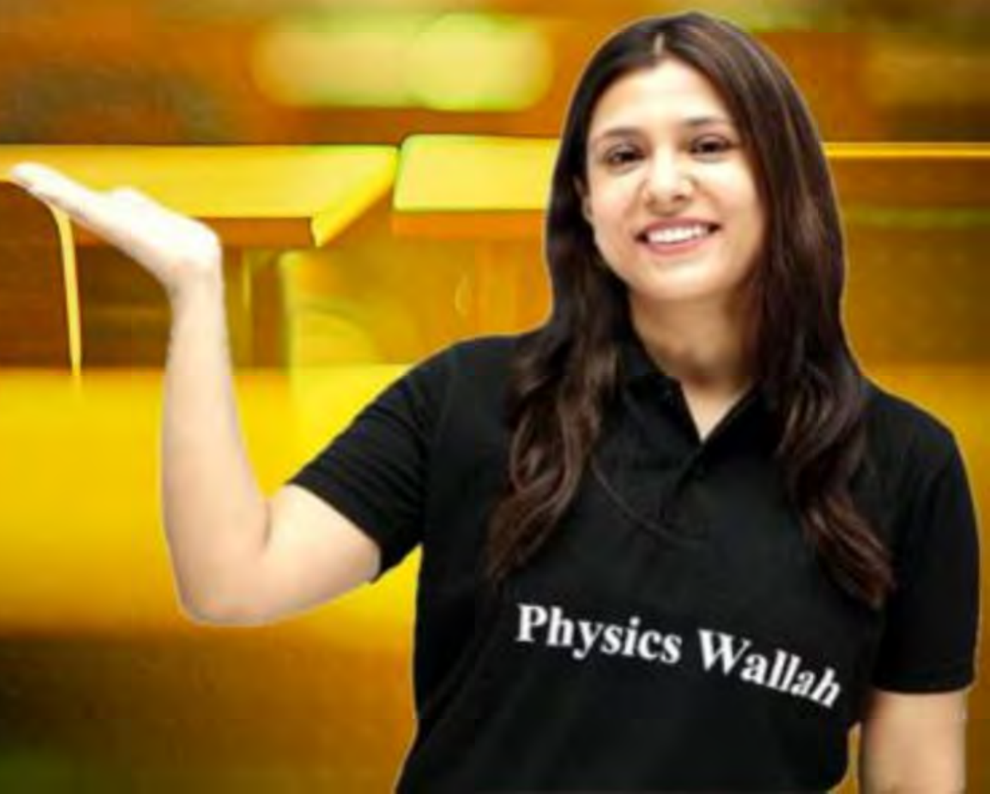
RADIANT 2026

Biology

The Flower

Lecture - 02

By - Upasana Mongia Ma'am



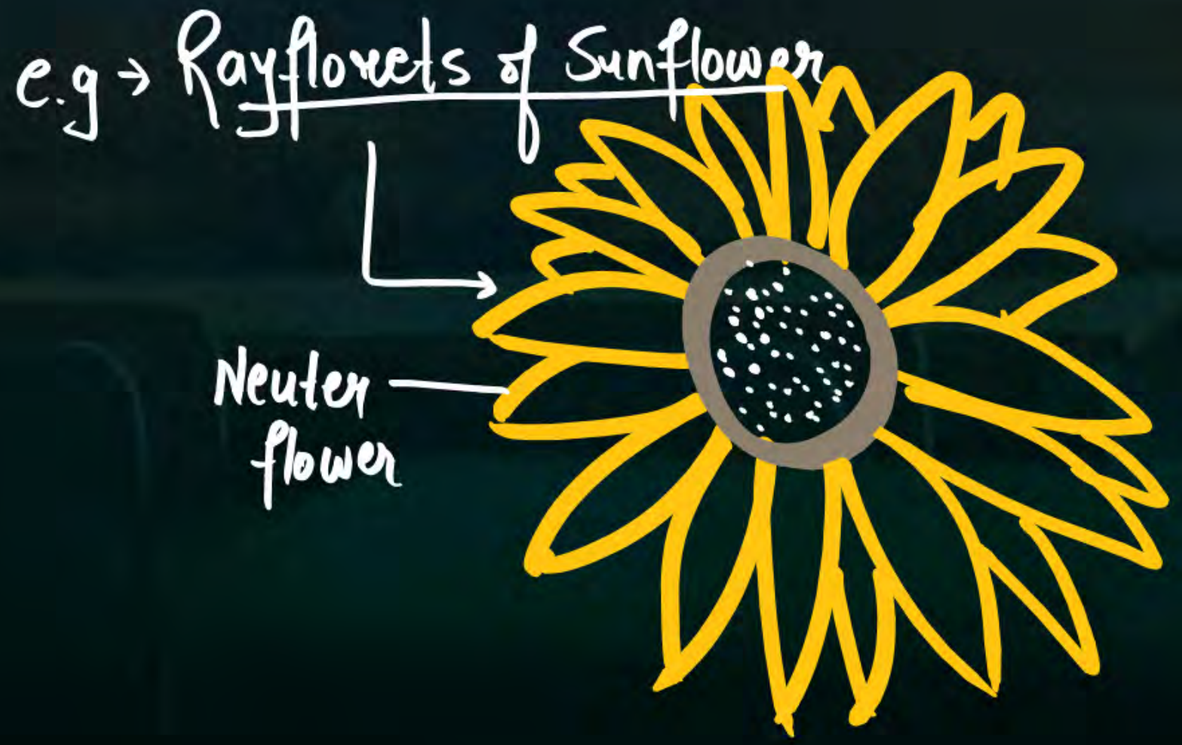
Topics *to be covered*

1. Functions of a flower



Neuter Flowers

The flowers which lack both Stamen and Carpel are called Neuter flower.



Tepals (Perianth)

- When petals and sepals cannot be differentiated from each other, then it is called tepals collectively called Perianth.
- Green-Sepaloid perianth
- Non-Green- Petaloid perianth



Tulip

Bracts

- A modified leaf or scale, typically small, with a flower or flower cluster in its axil.
- Bracts are sometimes larger and more brightly coloured than the true flower.

Modified leaf.



Bougainvillea

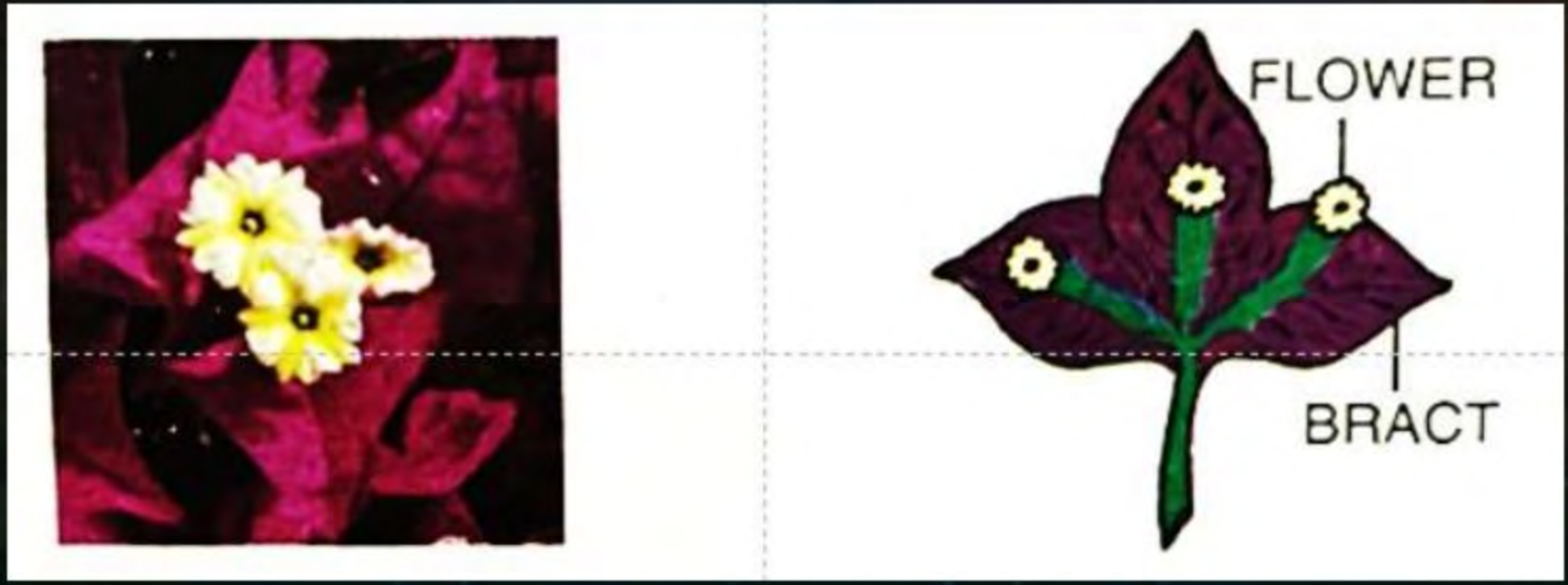
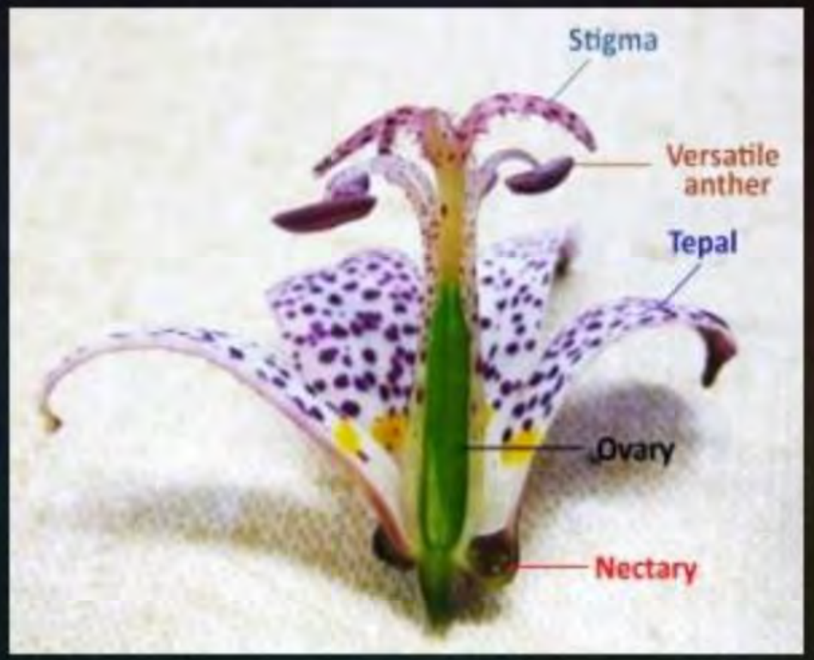


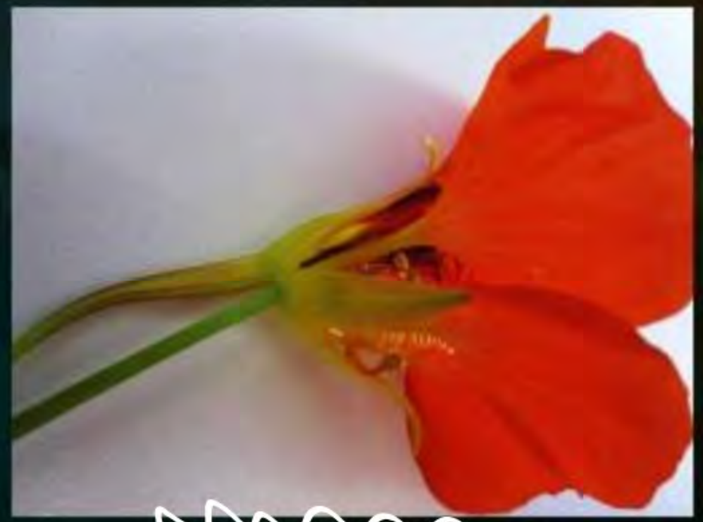
Figure: Bougainvillea, Large, coloured bracts bear the flowers

Nectaries

Nectaries:

- Most flowers produce a sweet fragrant liquid called nectar.
- Groups of nectar-secreting cells called nectaries are situated usually at the base of the pistil or on the bases of the petals.





Nasturtium

Doubts.

Sexuality in Flowers

♂♀

one house ♂♀

- BISEXUAL / HERMAPHRODITE/MONOECIOUS - A flower which contains both stamens and carpels e.g. hibiscus, rose lily and tomato, maize, cucumber, pumpkin

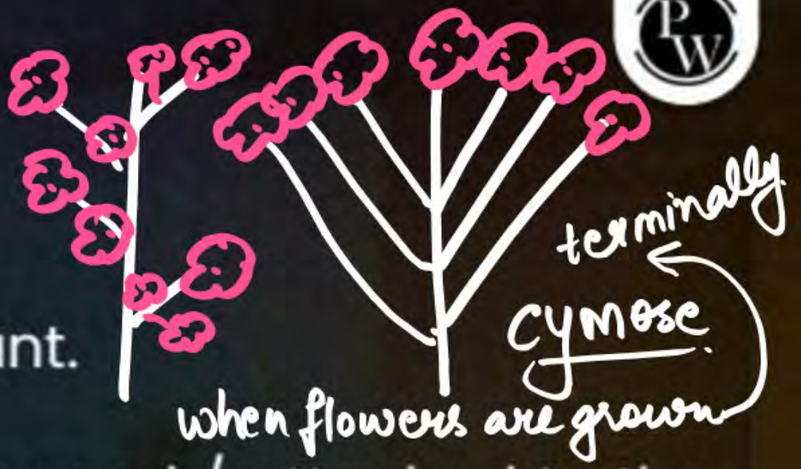
2 house

- UNISEXUAL / INCOMPLETE/DIOECIOUS - A flower which has only one of these parts, that is only the stamens or only the carpels, e.g. papaya and palm.

♂ : Stamen. Staminate

Pistillate

Inflorescence and Placentation



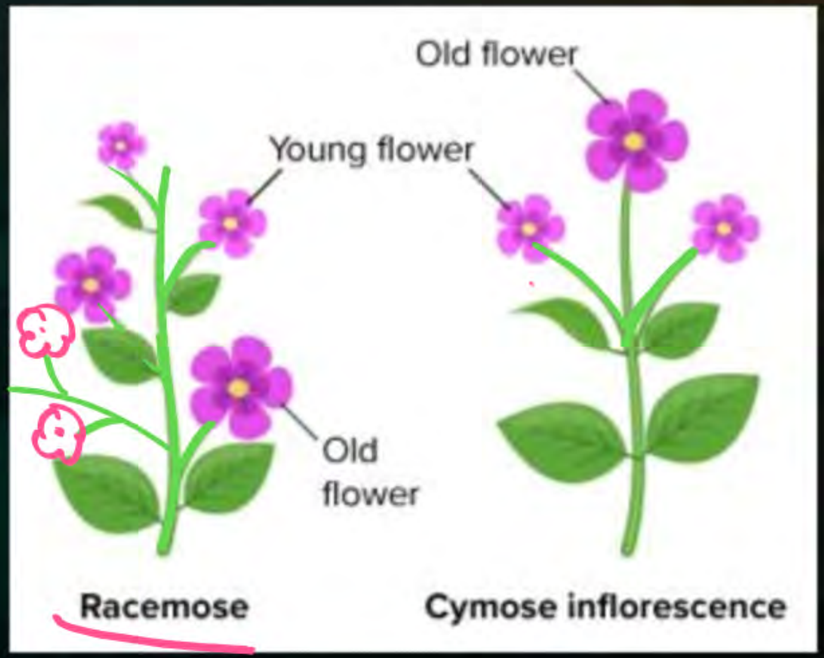
- Inflorescence - the arrangement of the flowers on a plant.
  Racemose → when the flowers are present axillary.
- Placentation is the manner in which the ovules are arranged / attached to the wall of the ovary.

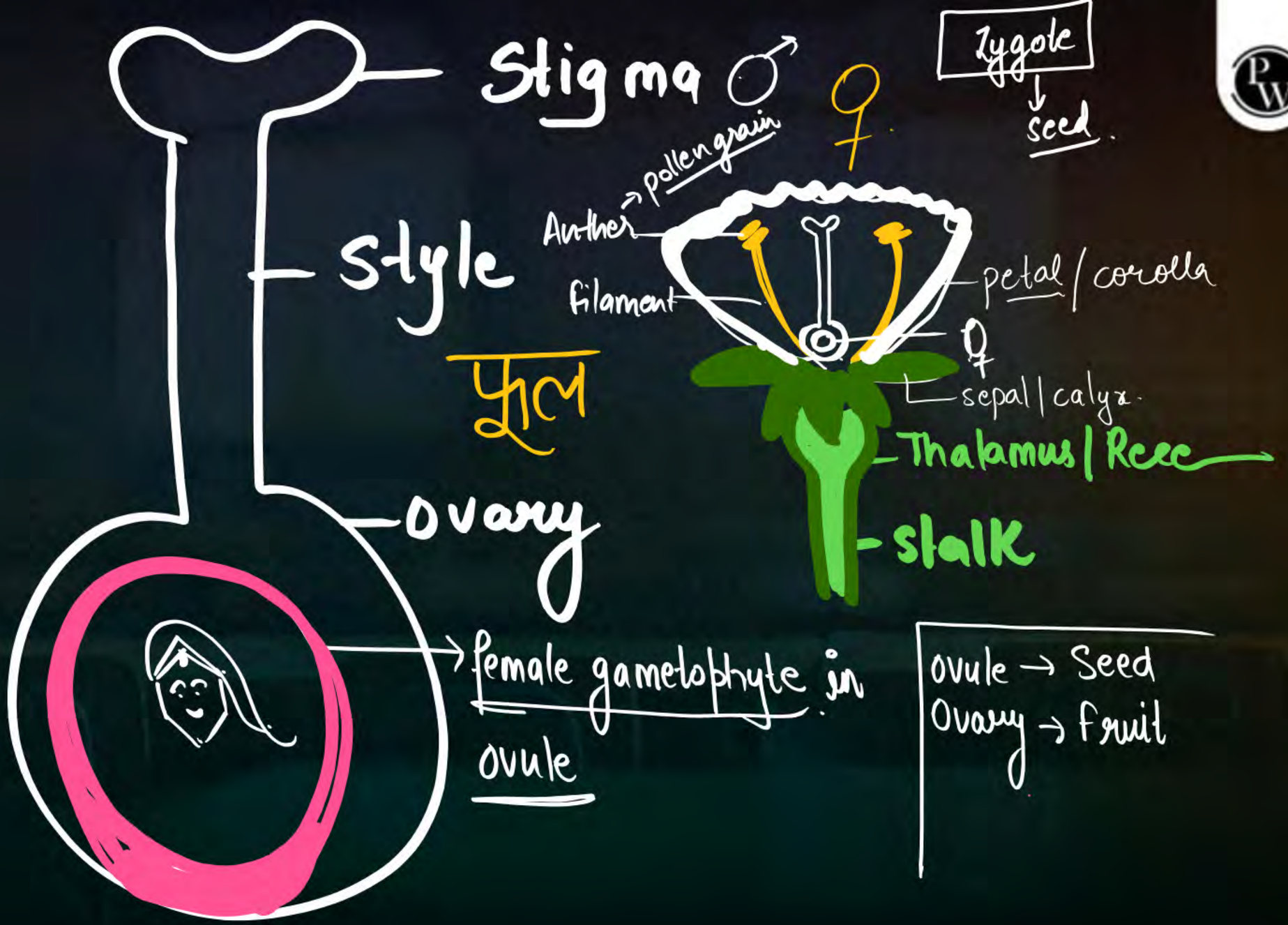
[cymose]

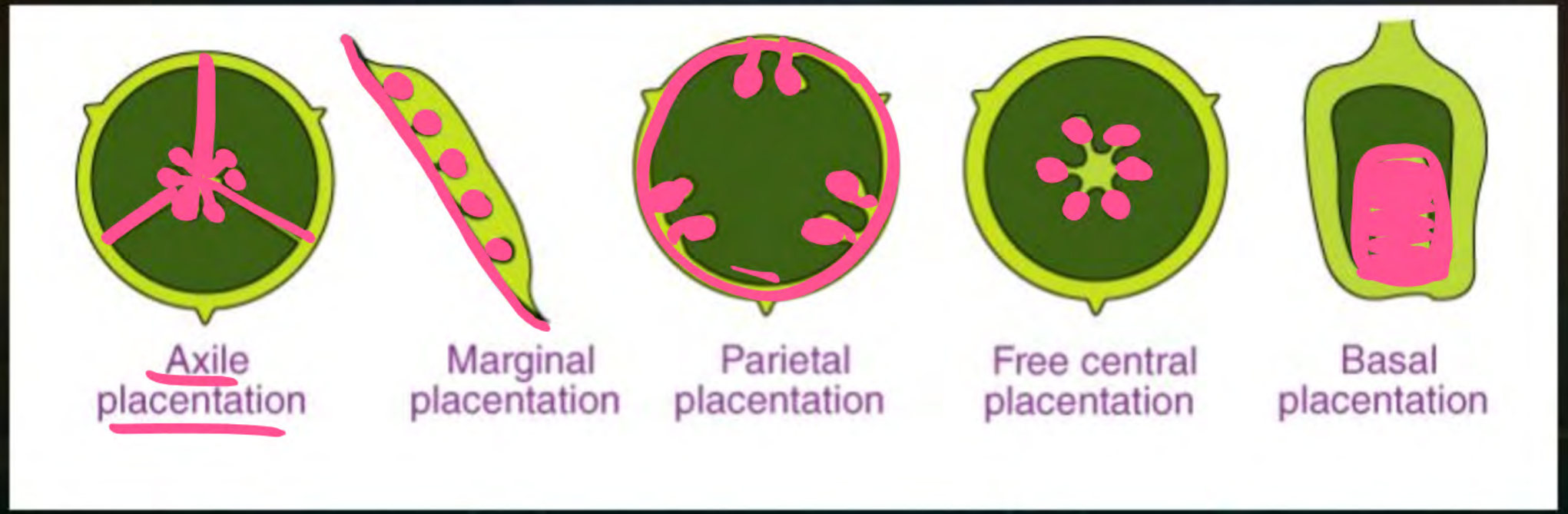


[Racemose]






Stigma
♂
Zygote
↓
seed.
Anther → pollen grain
♀
style
filament
petal / corolla
♀
sepal / calyx.
Thalamus / Rece
stalk
फूल
ovary
female gametophyte in ovule
ovule → Seed
Ovary → Fruit


Axile placentation
Marginal placentation
Parietal placentation
Free central placentation
Basal placentation

Question

The part of the flower that gives rise to the seed is

A Ovary

B Placenta

C Ovule

D Pollen grain

Question

Stigma may be simple or divided into two or more lobes.



A True

B False

Question

When a flower has both androecium and gynoecium

A. It is bisexual

B. It is unisexual

C. It is monoecious

D. It is dioecious

Question

2 group of brother

Which of the following shows diadelphous stamens?

A Solanum nigrum

potato

B Onion

C Pisum sativum

pea

D None of the above





Question

A bract is a modified leaf that is typically found just below a flower or flower cluster in plant.
Bracts can serve various functions, such as protecting the developing flower bud, enhancing the attraction of pollinators or providing structural support.
Which of the following part of flower is modified to make bract?

A Petal

B Sepal

C Leaf

D All of these

Question

The flowers which contain either only the pistil or only the stamens are called unisexual flowers. The flowers which contain both stamens and pistil are called bisexual flowers. The term neuter is given to a flower in which both:

(Specimen 2019)

A. Calyx and corolla are present

B. Calyx and corolla are absent

C. Androecium and Gynoecium are present

D. Androecium and Gynoecium are absent

Question

Corolla consists of coloured petals. They are normally large often has fragrance and bright coloured. Their primary function is to attract animals and insects for pollination. It can be free or tubular forming a tube [Free corolla are called:]

poly petal.

A Gamosepalous

B Gamopetalous

C Polysepalous

D Polypetalous

Question

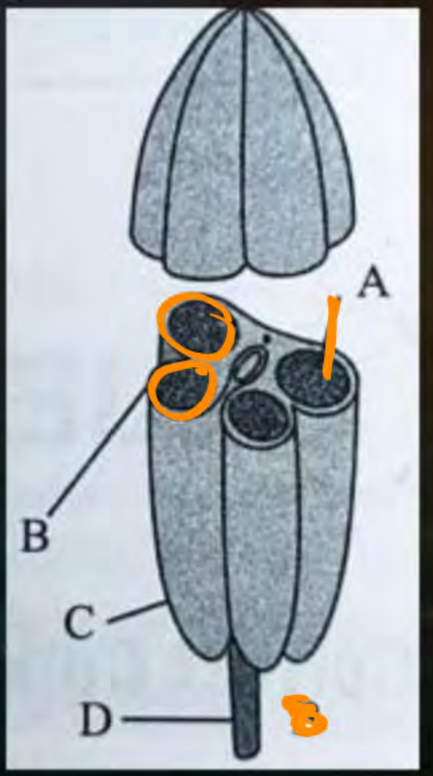
Study the figures carefully and answer the following questions:

(a) Identify the parts A–D.

(b) What does the figure represent?

↳ CROSS-SECTION of Anther.

A. Pollengrain
B Pollen Sac Bilobed
C Anther
D filament



Question

What type of androecium is found in China rose?

A Monoadelphalous

B Polyadelphalous

C Diadelphous

D Both (A) and (B)

Question

In the Questions below there are two statements marked as Assertion (A) and Reason (R): Read the statements given below and choose the correct option:

Assertion (A): Filaments consists of coloured petals. F. anthers.

Reason (R): Filament connects anther to filament. T.

A. Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

B. Both Assertion (A) and Reason (R) are correct but Reason (R) is not the correct explanation of Assertion (A).

C. Assertion.(A) is correct but Reason (R) is wrong.

D. Assertion (A) is wrong but Reason (R) is correct. ✓

Question

Assertion (A): Each carpel usually consist of two distinct parts.
Reason (R): Placentation refers to the mode of ovule attachment on the ovary wall.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(B) Both Assertion (A) and Reason (R) are correct but Reason (R) is not the correct explanation of Assertion (A).

(C) Assertion.(A) is correct but Reason (R) is wrong.

(D) Assertion (A) is wrong but Reason (R) is correct.

Question

Assertion (A): Calyx is inner whorl of flower which is green in colour.
Reason (R): Calyx help in photosynthesis.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(B) Both Assertion (A) and Reason (R) are correct but Reason (R) is not the correct explanation of Assertion (A).

(C) Assertion.(A) is correct but Reason (R) is wrong.

(D) Assertion (A) is wrong but Reason (R) is correct.

Question

Assertion (A): Bracts help in protecting the flowers.
Reason (R): Perianth protects reproductive parts.

Colourful
Green (photosynthesis) → Colourful
True.
group → Tepals [fused sepal & petal]

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(B) Both Assertion (A) and Reason (R) are correct but Reason (R) is not the correct explanation of Assertion (A).

(C) Assertion.(A) is correct but Reason (R) is wrong. ✓

(D) Assertion (A) is wrong but Reason (R) is correct.

Question

Fill in the blanks.

---------- the structure where the floral parts are borne in four whorls.

Thalamus.

Question

Epicalyx ---------- is the second series of sepals.



Question

The accessory whorls of the flower are ---calyx--- and corolla.

Question

Ovules are attached to the wall of the ovary by ----placenta----.

Question

_____Nectar.____ is the sweet fragrant liquid of flowers.

Question

True and False **(1 Mark Each)**

A flower can have either all whorls or not.

A True

B False

Question

There are six floral whorls present in a typical flower.

4

A True

B False

Question

Bracts can be large and colourful.

A True

B False

Question

Stamens and carpels are the male and female parts respectively.

A True

B False

Question

Pea is an example of diadelphous stamen.

A True

B False


Thank You